# tf.unique Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/unique

> *Finds unique elements in a 1-D tensor.*

## Function Signatures

```
tf.unique( x, out_idx=tf.dtypes.int32, name=None )
```

## Code Examples

```
tf.unique(
x,
out_idx=tf.dtypes.int32,
name=None
)
```

```
tf.unique(
x,
out_idx=tf.dtypes.int32,
name=None
)
```

```
tf.dtypes.int32
```

```
# tensor 'x' is [1, 1, 2, 4, 4, 4, 7, 8, 8]
y, idx = unique(x)
y ==> [1, 2, 4, 7, 8]
idx ==> [0, 0, 1, 2, 2, 2, 3, 4, 4]
```


```
# tensor 'x' is [1, 1, 2, 4, 4, 4, 7, 8, 8]
y, idx = unique(x)
y ==> [1, 2, 4, 7, 8]
idx ==> [0, 0, 1, 2, 2, 2, 3, 4, 4]
```


```
# tensor 'x' is [4, 5, 1, 2, 3, 3, 4, 5]
y, idx = unique(x)
y ==> [4, 5, 1, 2, 3]
idx ==> [0, 1, 2, 3, 4, 4, 0, 1]
```


```
# tensor 'x' is [4, 5, 1, 2, 3, 3, 4, 5]
y, idx = unique(x)
y ==> [4, 5, 1, 2, 3]
idx ==> [0, 1, 2, 3, 4, 4, 0, 1]
```

# Documentation

Finds unique elements in a 1-D tensor.

### View aliases

Compat aliases for migration

See
Migration guide for
more details.

tf.compat.v1.unique

### Used in the notebooks

• MoViNet for streaming action recognition

- Client-efficient large-model federated learning via `federated_select` and sparse aggregation

This operation returns a tensor y containing all of the unique elements of x
sorted in the same order that they occur in x; x does not need to be sorted.
This operation also returns a tensor idx the same size as x that contains
the index of each value of x in the unique output y. In other words:

y[idx[i]] = x[i] for i in [0, 1,...,rank(x) - 1]

### Examples:

## Returns